Release Plan v2: Automated Timetable & Student Portal

Written: Saturday 7 November 2026 · Hand-in: Sunday 15 November 2026 · Plan owner: Sajawal

What is new in v2

- The two code reviews (backend and frontend) are merged into this plan. **This is now the only document the team reads.**
- Taha and Eisha have swapped roles. **Taha is now frontend. Eisha is now backend.**
- The work is divided equally by hours, and v1's capacity mistake is fixed (Part 5.2).
- Every change is listed in the change log at the end.

How to read this

- **Everyone:** read Parts 1 to 6 (about 15 minutes), then **your own page in Part 7**. Your page alone tells you what to do each morning.
- **Sajawal:** also read the Appendix (technical detail and the full list of review items). The team part does not depend on it.

Part 1. The short version

1. **Can we finish by 15 November? Yes, but only if we build much less than the proposal and the two reviews ask for.**
2. All work, costed task by task, comes to **117 hours**. Before the 14 November freeze we have **78 hours** of feature time. **We are 39 hours short.**
3. **What we keep (74 hours):** the timetable system working on the internet. That means login for all 4 roles, each person's timetable, swaps with HOD approval and a clash check, makeup classes, notifications, sample data, and the critical review fixes.
4. **What waits:** the portal screens, several review clean-ups, and some safety extras. All of them are listed in Part 5.4 with the reason and who explains each to the instructor.
5. **Biggest risks:**
    - **Eisha (now backend) has no commits yet** and owns the swap fix and the first deploy.
    - **The frontend has almost no spare time.**
    - **The code has never run on MySQL.**

What this plan assumes

- Each person works **3 hours a day**, from 8 to 15 November. All five members are available (you told me).
- About **0.4 hours per person per day** goes to check-ins and reviews. That leaves **2.6 hours of task work per day**.
- **14 November is freeze day:** bug fixes only, no features. **15 November is for handing in**, plus emergencies.
- The frontend stays as **plain HTML and JavaScript** (React is not required).
- Estimates are mine, made by reading the code. Nothing is padded in secret; all estimates are in the Appendix.
- I could not check whether the free hosting plans have changed. That is checked on 9 November.

Part 2. Glossary (words used in this plan)

Term	What it means	Why it matters to us
Repo (repository)	The project folder on GitHub with all our code and its history.	It is the single source of truth. Code sent on WhatsApp doesn't count.
Branch	Your own copy of the code, where you work without disturbing others.	You never work directly on `main`.
main	The official branch. Only reviewed work goes in.	What is on `main` is what goes online.
PR (pull request)	A request to add your branch's changes to `main`.	It is the only way code reaches `main`.
Review / approval	A teammate reads your PR, runs it, and clicks Approve.	`main` is locked until one other person approves.
Squash merge	Your PR goes into `main` as one single change.	It keeps history clean. Delete your branch afterwards.
Merge conflict	Two people changed the same lines and Git can't choose.	This is why some files have one owner (Rule 4).
Backend	The server code (Python, FastAPI) that stores data and enforces rules.	It is built by Sajawal, Wajiha and Eisha.
Frontend	The web pages people click on (HTML, CSS, JavaScript).	It is built by Amna and Taha.
API	The web addresses the frontend calls to get or send data.	It connects frontend and backend.
Endpoint	One API address, for example `POST /api/auth/login`.	Each screen uses a few of these.
API contract	One document listing every endpoint: address, what you send, what you get back.	Both sides build to the same document.
Mock data	Fake sample data in files, shaped exactly like real API answers.	Frontend work can start before the backend is ready.
FastAPI	The Python tool our backend is built with.	It gives us `/docs` for free.
/docs	An automatic web page where you can try every endpoint.	Testing without the frontend, and showing portal features in the demo.
Postman	An app for sending test requests to the API.	Saved example requests.
JWT (token)	A signed "ticket" the server gives you at login. The browser sends it with every request.	It is how the server knows who you are.
Role	Student, teacher, HOD or admin.	It decides what each person may do.
HOD	Head of Department.	Approves swaps for their own department.
Swap	A teacher asks to move a class to another time or room.	Core feature; needs HOD approval.
Makeup class	An extra class booked to replace a missed one.	Core feature; the system finds free times and rooms.
Conflict check	Code that checks if a teacher, section or room is already busy.	Stops double-bookings.
Row lock	While one person saves a booking, the database makes others wait.	Stops two people booking the same room at once.
Database / MySQL	Where all data is stored. MySQL is the type we use online.	Data must live somewhere permanent.

Term	What it means	Why it matters to us
SQLite	A small test database.	Our tests use it. It behaves differently from MySQL, so tests passing on it don't prove the online version works.
Migration / Alembic	Alembic is a tool that creates database tables from our code, step by step.	Everyone, and the server, gets the same tables.
Seed data	Realistic sample data: rooms, teachers, a timetable, test accounts.	Without it every screen is empty.
Test / pytest	Code that checks other code automatically. `pytest` runs all tests.	Catches bugs before they reach `main`.
CI	GitHub runs all tests automatically on every PR.	A PR with failing tests can't be merged.
PR template	A checklist GitHub adds to every new PR.	Nobody forgets "tests pass, no secrets".
Deploy	Putting the app on a public server anyone can open.	"Works on my laptop" isn't enough to hand in.
Staging	The online copy where we test before handing in.	Finds "only breaks online" bugs early.
Render / Aiven / Netlify	Free hosting websites: Render runs the backend, Aiven the MySQL database, Netlify the frontend pages.	Where our app lives online.
TiDB	Another free database host that behaves like MySQL.	Backup if Aiven fails.
Environment variable / `.env`	A setting, such as a password, given to the app from outside the code.	Passwords never go in the repo.
Secret	Any password or key.	If committed to Git, it is leaked forever.
CORS	A browser safety rule: the backend must list which websites may call it.	If wrong, it works locally but fails online.
Cold start	Free Render servers sleep when unused and take 30–60 seconds to wake.	Open the site 2 minutes before the demo.
Temporary disk	Files saved on a free Render server vanish on restart.	Don't demo uploads unless done live.
Health check	`/api/health`: an endpoint that answers "OK".	Proves the server is up; wakes it before the demo.
Status codes	401 not logged in · 403 not allowed · 404 not found · 409 clash or duplicate · 500 server crashed.	The frontend shows a message for each. A 500 is always a bug.
Smoke test	A quick run through the main steps after a deploy.	Catches big breakages fast.
Person-hour (h)	One person working one hour.	The unit of all our maths.
Capacity	Total hours the team can really work.	It decides how much we can build.
Buffer	Spare hours kept for surprises.	Things take longer than planned.
Estimate (best / likely / worst)	Three guesses per task.	We plan on "likely" and watch "worst".
Task list (WBS)	The full list of tasks. WBS = Work Breakdown Structure.	Small tasks are easy to track and hand over.
Dependency	Task B can't start until task A is done.	Waiting is where time is lost.
Critical path	The chain of dependent tasks that decides the finish date.	If any of them is late, the project is late.

Term	What it means	Why it matters to us
MoSCoW	Sorting work into **M**ust, **S**hould, **C**ould and **W**on't have this time.	It decides what gets dropped first.
Cut line	The line between what we build and what we drop.	It is drawn so the Musts fit our hours.
Feature freeze	After this date, only bug fixes. Ours is **14 November**.	It protects a working demo.
Trigger	A rule agreed in advance: "if X is late by this time, we automatically do Y".	No arguing when we are behind.
Review time limit	Every PR is reviewed within 4 waking hours.	Slow reviews block the next person.
Review item	One problem found in the backend or frontend code review.	Every one of them is now a task or is listed as deferred.
Traceability table	A table showing where each review item went.	Proves nothing from the reviews was lost.

Part 3. Code legend (one scheme for everything)

Every code below is always written together with its plain name.

Prefixes

Prefix	Means	Example
BE-nn	Backend task	BE-01: Make the backend start
FE-nn	Frontend task	FE-04: Timetable page
OP-nn	Operations task (tests on GitHub, deploy, docs)	OP-02: First deploy
BR-xx	An item from the **B**ackend code **R**eview. Used only to show where a task came from.	BR-C6: swaps use a column that doesn't exist
FR-xx	An item from the **F**rontend code **R**eview. Same use.	FR-C2: login accepts any password
TR-n	Trigger (automatic rule)	TR-2: early swap check
RK-n	Risk	RK-1: Eisha stalls

In BR- and FR- codes, the letter after the dash is the review's severity: **C** = critical, **M** = major, **D** = moderate, **N** = minor. The reviews' own short codes (C1, M3, D2…) are not used in this plan.

All tasks (priority: Must, Should, Could, Won't; hours are "likely")

Code	Task name	Owner	Hours	Priority
BE-01	Make the backend start	Sajawal	3	Must
BE-02	Fix broken import paths and package files	Wajiha	1.5	Must
BE-03	Fix events update and instructor name	Wajiha	1.5	Must
BE-04	Fix login and user accounts	Sajawal	4	Must
BE-05	Write the single API contract	Wajiha	3	Must

Code	Task name	Owner	Hours	Priority
BE-06	Fix the swap feature and conflict checker	Eisha	5	Must
BE-07	Let each HOD approve their own department's swaps	Eisha	3	Must
BE-08	Make makeup booking safe	Sajawal	4	Must
BE-09	Test notifications	Eisha	2	Must
BE-10	Create database tables on MySQL (migrations)	Wajiha	3	Must
BE-11	Fill the database with sample data	Wajiha	4	Must
BE-12	Limit login attempts	Sajawal	1	Should
BE-13	Use the real roll-number format	Wajiha	0.5	Should
BE-14	Check uploaded files by content	Eisha	1	Should
BE-15	Swaps also check against booked makeup classes	Eisha	3	Could
BE-16	One role-check system; remove stale comments	Eisha	2.5	Could
BE-17	Safer database transactions in services	Eisha	2	Could
BE-18	Unknown room capacity instead of a fake 40	Wajiha	1	Could
BE-19	Read the current term from settings	Sajawal	1	Could
BE-20	Backend tidy-up (dev tools file, folder rename, names out of files, ownership file, move the .docx, style checker)	Wajiha	3.25	Could
BE-21	Import the real AI/CS/IT timetable sheets	Wajiha	3	Could
FE-01	Clean up the frontend and bring it into `main`	Amna	3	Must
FE-02	Small JavaScript fixes	Taha	0.75	Must
FE-03	Real login on the frontend	Taha	6.5	Must
FE-04	Timetable page	Amna	6	Must
FE-05	Swap requests page	Amna	6	Must
FE-06	Makeup class page	Taha	5	Must
FE-07	Notification bell	Taha	3	Must

Code	Task name	Owner	Hours	Priority
FE-08	Secure software link and faster hero image	Amna	0.35	Should
FE-09	Sensible year list on the lectures page	Taha	0.25	Should
FE-10	One shared menu/layout on every page	Taha	2	Could
FE-11	Activate account page	Taha	1.5	Could
FE-12	Empty rooms page	Amna	3	Could
FE-13	Past papers and lecture slides on real data, with real uploads	Taha	5	Could
FE-14	GPA page on real data	Taha	3	Could
FE-15	Software list and events pages on real data	Taha	3	Could
FE-16	One font and a code style checker	Amna	1.5	Could
FE-17	Accessibility pass	Amna	1.5	Could
OP-01	Automatic tests on every PR (CI) and a PR template	Eisha	2	Must
OP-02	First deploy: "hello world" online	Eisha 2.5 + Sajawal 0.5	3	Must
OP-03	Full deploy with real tables and sample data	Sajawal	3	Must
OP-04	Write the README (run, deploy, scope decision)	Wajiha	2	Must
OP-05	Postman requests for the demo	Eisha	1.5	Could
OP-06	Run CI tests on MySQL too	Eisha	2	Could
n/a	Password reset, live push notifications, React	n/a	n/a	Won't

Totals: **Must 74.25 h · Should 3.1 h · Could 39.75 h · all work 117.1 h.**

Triggers:

- **TR-1:** backend not starting.
- **TR-2:** early swap check.
- **TR-3:** swap tests not green.
- **TR-4:** frontend still on fake data.
- **TR-5:** MySQL not working.
- **TR-6:** member silent for 2 days.
- **TR-7:** demo failing on staging.
- **TR-8:** Amna behind.

All are explained in Rule 6.

Risks: RK-1 to RK-10, explained in Part 6.

Part 4. The rules, in plain words

Rule 1. What "done" means (one definition for everyone)

A task is **done** only when all four are true:

1. It is **merged into** `main` through an approved PR.
2. The **automatic tests are green**. Backend work also has its own test.
3. **It works on the online staging site**, not only on your laptop. (Until staging exists on 9 November, "works locally" is accepted.)
4. If it changed an endpoint, the **API contract is updated**.

"I pushed it", "it works on my machine" and "the PR is open" all mean **not done**.

Rule 2. Why feature freeze is 14 November

Every new feature can break something that already worked. The demo must run **twice in a row without errors** before we hand in. If we add features until the last day, there's no time left to fix what they break. So from 14 November we only fix bugs and rehearse. 15 November stays empty on purpose as a safety day. **This also means all feature work must finish by 13 November.** That is why each person has 15.6 hours for tasks, not 18.2 (Part 5.2).

Rule 3. Why a review must happen within 4 waking hours (not 24)

Four key tasks happen **one after another**:

1. BE-01: Make the backend start
2. BE-04: Fix login and user accounts
3. BE-10: Create database tables on MySQL
4. BE-11: Fill the database with sample data

Each one waits for a review. At 24 hours per review we could lose **4 of our 6 feature days**. With 4 hours, a PR opened in the morning is merged the same day.

- Every PR has a **main reviewer** and a **backup reviewer** (table below). If the main reviewer is silent for 4 hours, the backup reviews.
- **Backend PRs are reviewed by a backend member. Frontend PRs are reviewed by a frontend member.** With only two frontend members, Sajawal is the frontend backup reviewer. That is the one exception, so that a frontend PR never waits more than 4 hours.
- The reviewer **runs the code**, not only reads it. The bugs on `main` today got in because nobody ran the code (from BR-C7: code was merged without ever being run).
- Keep PRs small (about 300 changed lines or less).
- The PR title must say what the PR really does. "Integration done" on a PR with no backend calls is not allowed (from FR-C4: zero backend integration despite the commit title).

Whose PR	Main reviewer	Backup reviewer
Sajawal (backend)	Wajiha	Eisha
Wajiha (backend)	Sajawal	Eisha
Eisha (backend)	Sajawal for BE-06 and OP-02 (he wrote the swap code); Wajiha for the rest	the other one
Amna (frontend)	Taha	Sajawal
Taha (frontend)	Amna	Sajawal

Whose PR	Main reviewer	Backup reviewer
BE-05 API contract	Sajawal **and** Taha (one backend + one frontend, as the guide requires)	n/a

Rule 4. One owner per shared file

Only the owner edits these files. Others ask the owner in their PR.

- `backend/app/main.py` and `backend/requirements.txt`: **Sajawal**
- `frontend/js/common.js` and `frontend/js/api.js`: **Taha**. This starts once Amna's FE-01 (frontend cleanup) is merged on 8 November, because FE-01 has to fix `common.js` first.
- Database migration files: **Wajiha**

Rule 5. Why the cut line is where it is

- Before the freeze we have **78 hours** of task time. The Musts need **74.25 hours**. That leaves 3.75 hours, which holds 3.1 hours of small Should tasks.
- The Musts are what the project's title promises: the automated **timetable**, swaps, makeup classes, notifications, all online. They also include **every critical review fix**, because the app doesn't work without them.
- Portal screens were cut first because their backend **already works and has tests**. They can still be shown through `/docs`.
- The frontend decided most cuts. Amna and Taha together have 31.2 hours before the freeze and the frontend Musts need 30.25. No frontend Should or Could task fits.
- **Must before Should:** nobody starts a Should task until all their Must tasks are done. If any Must runs late, Should tasks are dropped first.

Rule 6. What happens automatically if we fall behind (triggers)

These rules are decided **now**, so nobody argues later. At the 21:00 check-in, Sajawal says which trigger fired, and it happens.

Trigger	If this is not done by…	…then this happens automatically
TR-1: backend not starting	BE-01 (make the backend start) by **8 Nov 23:59**	All Should tasks are cancelled. Wajiha writes BE-05 (API contract) from the existing code instead of waiting.
TR-2: early swap check (new)	Eisha has pushed a branch for BE-06 (fix the swap feature) with the column rename by **9 Nov 21:00**	Sajawal takes BE-06 at once; he wrote this code. In exchange, Eisha takes OP-03 (full deploy) and the staging smoke test. BE-12 (limit login attempts) is cancelled.
TR-3: swap tests not green	BE-06 (fix the swap feature) merged with green tests by **10 Nov 21:00**	The same exchange as TR-2, if it has not already happened.
TR-4: frontend still on fake data	Every finished page uses the real API by **12 Nov 21:00**	FE-06 (makeup page) shows only "find free slots"; booking is shown through `/docs`. All frontend Should tasks are cancelled.
TR-5: MySQL not working	Tables created on online MySQL by **10 Nov 21:00**	Switch to TiDB. If that also fails by 11 Nov, switch to Postgres (this would have to be explained to the instructor).
TR-6: member silent	Anyone has **no PR for 2 days in a row**	Their remaining work goes that night to their named backup (see their page).
TR-7: demo failing on staging	The full demo passes on staging by **13 Nov 21:00**	No more features. 14 Nov is spent only on making the demo pass.

Trigger	If this is not done by…	…then this happens automatically
TR-8: Amna behind	FE-04 (timetable page) merged by **11 Nov 21:00**	Taha takes the HOD half of FE-05 (swap requests page, about 1 hour). To pay for it, FE-07 (notification bell) becomes an unread-count badge only, with no dropdown, and FE-09 (year list) is cancelled.

Rule 7. Daily check-in (5 minutes, WhatsApp, 21:00, one message each)

```
[Name] Day N
Done (merged PR numbers): …
Blocked by: … (who or what)
Tomorrow (task codes + names): …
My Must hours: X of Y done · plan said Z · ahead / on track / behind
```

Sajawal then posts one line: "Critical path on track / behind by N hours · Trigger fired: none / TR-#".

Part 5. Where we are and what we have

5.1 Where the project is today (checked on 7 Nov)

- **The backend does not start.** The startup file (`main.py`) is empty and three files it needs don't exist. When I ran the tests, **0 tests ran**.
- **Much backend code exists and most of it is good:** about 5,200 lines and 53 endpoints. The backend review found 112 of 116 tests pass once the loading problems are patched. Most backend work is **repair**.
- **Broken backend features:**
  - Login: the code and the user table disagree.
  - Swaps: the code uses a column name that doesn't exist.
  - Makeup booking: lecture courses get put into labs, and nothing stops double bookings.
- **The frontend exists only as portal pages, with fake data.** It is on the `frontend` branch; its PR was closed without merging on 6 Nov.
  - Login accepts any password.
  - The signup and lecture pages crash.
  - **No timetable screens exist.**
- **There is no single API contract.** Login, swaps, makeup and notifications have none.
- **Missing:** automatic tests, a PR template, database setup, sample data, deployment and a README.
- **Good:** no secrets are in the repo.
- **Activity:** no commits from anyone since 24 October.

5.2 The hours (capacity check), corrected

	Hours
5 people × 3 hours × 8 days (8–15 Nov)	120
minus 15 November (hand-in and emergencies)	−15
minus check-ins and reviews (0.4 h × 7 days × 5 people)	−14
Real working hours, 8–14 Nov	**91** (18.2 per person)
minus 14 November (freeze: bug fixes only, 2.6 h × 5)	−13
Feature hours before the freeze, 8–13 Nov	**78** (15.6 per person)

	Hours
Must work	**74.25**
Should work	3.1
Spare before the freeze	0.65
Buffer after the freeze (14 Nov bug fixing + 15 Nov reserve)	13 + 15 = 28 hours (38% of the Must hours)
All costed work (Must + Should + Could)	**117.1**
Shortfall against feature hours	**39.1 hours**

A correction to v1, stated plainly: v1 treated all 91 hours as feature time. 14 November is freeze day, so feature work must end on 13 November. v1 had Amna at 17 hours of Musts, which could not have finished before the freeze. v2 fixes this.

5.3 Equal division of the work (Must hours, "likely")

"Before" means the v1 plan with Taha and Eisha's names swapped. The team average after is 14.85 hours.

Person	Before	After	Difference from average	After + Should	Limit before freeze
Sajawal	16.5	14.5	−0.35	15.5	15.6
Wajiha	13.0	15.0	+0.15	15.5	15.6
Eisha	16.0	14.5	−0.35	15.5	15.6
Amna	15.0	15.0	+0.15	15.35	15.6
Taha	13.0	15.25	+0.40	15.5	15.6
Team	**73.5**	**74.25**	everyone within 0.4 h	**77.35**	**78**

What moved to make this equal:

- **Sajawal −2 hours:** the README (OP-04) went to Wajiha, and his share of the first deploy (OP-02) dropped from 1 hour to 0.5. Sajawal is not the dumping ground for review fixes. His only review fixes are in code he wrote himself.
- **Taha:** took FE-06 (makeup page), because it is frontend work, and FE-02 (small JavaScript fixes) in files he co-wrote.
- **Eisha:** took OP-01 (automatic tests on GitHub), which is backend tooling, and kept the backend swap, HOD and notification work.
- **FE-10 (shared layout) became Could.** Once FE-01 loads `common.js` on every page, its existing menu appears everywhere, which is enough for the demo.

5.4 What we build and what we cut

- **Must (74.25 h):** BE-01 to BE-11, FE-01 to FE-07, OP-01 to OP-04.
- **Should (3.1 h), on 13 November, only if your Musts are merged:**
  - BE-12 (limit login attempts)
  - BE-13 (real roll-number format)
  - BE-14 (check uploaded files by content)
  - FE-08 (secure software link and faster image)
  - FE-09 (sensible year list)
- **Could (39.75 h), deferred until after 15 November.** None of them fits before the freeze:

Deferred task	Hours	Why deferred	Must be explained to the instructor? Who explains
BE-15: Swaps also check against booked makeup classes	3	No hours left; the demo avoids this case	**Yes**, the conflict check is only partial. Eisha
BE-16: One role-check system; remove stale comments	2.5	Internal tidy-up; the app works without it	No
BE-17: Safer database transactions	2	Internal; works correctly today	No
BE-18: Unknown room capacity instead of a fake 40	1	The sample data uses real capacities	No
BE-19: Read the current term from settings	1	One term is enough for the demo	**Yes**, only one term is supported. Sajawal
BE-20: Backend tidy-up	3.25	No effect on users	No
BE-21: Real AI/CS/IT timetable sheets	3	No hours; the SE format works	**Yes**, SE format only. Wajiha
OP-05: Postman requests for the demo	1.5	`/docs` covers the same need	No
OP-06: CI tests on MySQL	2	MySQL is tested by hand on staging instead	No
FE-10: One shared menu/layout	2	The `common.js` menu is enough	Maybe (pages look different). Taha
FE-11: Activate account page	1.5	Demo accounts already have passwords	**Yes**, activation works only through `/docs`. Sajawal
FE-12: Empty rooms page	3	No frontend hours	**Yes**, the API works but there's no screen. Amna
FE-13: Past papers and slides on real data	5	No frontend hours	**Yes**, the portal screens use fake data. Taha (screens), Wajiha (APIs)
FE-14: GPA page on real data	3	No frontend hours	**Yes**, same as above. Taha
FE-15: Software and events pages on real data	3	No frontend hours	**Yes**, same as above. Taha
FE-16: One font and a style checker	1.5	Looks only	No
FE-17: Accessibility pass	1.5	No hours	Maybe. Amna

- **Won't have this time:**
  - A real password reset. The page will say "contact your department admin".
  - Live push notifications. We check every 15 seconds instead, which the proposal allows.
  - React. Plain HTML/JS instead; Amna explains this to the instructor.
- **Also to explain to the instructor:**
  - Students can't sign up freely; they use accounts the admin created (Sajawal).
  - Notifications refresh every 15 seconds (Eisha).
  - If TR-5 fires, a database other than MySQL (Sajawal).

5.5 The critical path (the chain that decides our finish date)

1. **BE-01** Make the backend start (Sajawal, 8 Nov) →
2. **BE-04** Fix login and user accounts (Sajawal, merged by 13:00 on 10 Nov) →
3. **BE-10** Create database tables on MySQL (Wajiha, 10 Nov) →
4. **BE-11** Fill the database with sample data (Wajiha, core data by 18:00 on 11 Nov) →
5. **FE-04** Timetable page on real data (Amna, 11 Nov) →

6. **FE-05** Swap requests page (Amna, 12–13 Nov) →
7. **OP-03** Full deploy (Sajawal, 12 Nov) → full demo passes on staging (13 Nov) →
8. Feature freeze (14 Nov).

Running alongside it, and also required:

- **FE-03** Real login on the frontend (Taha, merged 10 Nov)
- **BE-06** Fix the swap feature (Eisha, merged 10 Nov)

5.6 Do the milestones still hold?

Milestone	Holds?
API contract final, 9 Nov	**Yes.**
First deploy online, 9 Nov	**Yes on paper**, but it is owned by Eisha, who has no commits yet (RK-1).
Frontend on real APIs, 12 Nov	**Only partly.** No page uses fake data after 12 Nov. But two pieces merge on **13 Nov**: the HOD half of FE-05 (swap requests page, 1 h) and FE-07 (notification bell, 2 h). The frontend has no hours to finish them earlier.
Staging deployed and demo passing, 13 Nov	**Yes**, with about 0.1 hours of frontend slack. Any frontend slip fires TR-7 or TR-8.
Feature freeze, 14 Nov	**Yes.**

Part 6. Risks (what could go wrong)

Track records checked against git history and the GitHub collaborator list on 7 Nov.

#	Risk	How likely	How bad	Early warning sign	What we do
RK-1	**Eisha (now backend) stops delivering.** She has write access as `Eishashahzad` but **0 commits, 0 PRs and 0 tests** on any branch. She owns BE-06 (fix the swap feature, on the critical path) and OP-02 (first deploy, due 9 Nov).	High	High	No BE-06 branch pushed by 9 Nov 21:00	TR-2 (early check). Backup: **Sajawal** for BE-06 and OP-02 (he wrote the swap code in PR #33 and leads deploy). **Wajiha** for BE-07, BE-09 and OP-01.
RK-2	**Code breaks on MySQL.** It has only ever run on the SQLite test database.	High	High	Table creation fails online on 10 Nov	Real database on day 3. TR-5.
RK-3	**Merge conflicts** on shared files	Medium	Medium	Two open PRs change the same file	One owner per shared file (Rule 4).
RK-4	**Slow reviews**	High	High	A PR waits more than 4 hours	4-hour limit, backup reviewer, small PRs (Rule 3).
RK-5	**Real timetable sheets or roll numbers don't match the code**	Medium	Medium	Test import of the real SE sheet shows errors	Sample data comes from a file we control. BE-13 (real roll-number format) is a Should; BE-21 (real sheets) is deferred.

#	Risk	How likely	How bad	Early warning sign	What we do
RK-6	**Double booking** when two people book at once	Medium	Medium	In a two-tab test, both bookings succeed	BE-08 (safe makeup booking) adds a lock and a database rule. Eisha runs a two-tab test on staging on 13 Nov.
RK-7	**Works locally, breaks online**	High	High	Something works on a laptop but fails on the website	Online from 9 Nov; redeploy after every merge; smoke test; wake the server before the demo.
RK-8	**Polishing portal pages instead of timetable pages**	Medium	High	Portal commits while FE-04 and FE-05 aren't merged	Must before Should (Rule 5).
RK-9	**Taha's real login (FE-03) is harder than expected.** He has 2 commits on the `frontend` branch (21 and 24 Oct) and is co-author on 2 more, but **0 tests and 0 API calls**. His earlier login code is exactly the fake login the review flagged (from FR-C2: login accepts any password). FE-03 blocks every other frontend page.	Medium	High	FE-03 not calling the real login endpoint by 10 Nov evening	Backup: **Sajawal** (he wrote the backend login). Amna has no spare time to take it.
RK-10	**The frontend has almost no spare time.** Before the freeze, Amna has 0.25 h spare and Taha 0.1 h.	Medium	High	Either is more than 1 hour behind at a check-in	Should tasks are dropped first, then TR-8, then TR-4.

6b. If Amna falls behind: what gets cut first

Amna owns FE-04 (timetable page) and FE-05 (swap requests page), both on the critical path.

1. Her Should task FE-08 (secure link and faster image) is dropped.
2. **TR-8:** if FE-04 is not merged by 11 Nov 21:00, **Taha** takes the HOD half of FE-05. FE-07 (notification bell) becomes a simple unread-count badge to pay for it.
3. If the teacher half of FE-05 isn't merged by 12 Nov 21:00, the timetable becomes a **simple table per day** and teachers pick the class to swap from a **dropdown**.
4. **Never cut:** teacher asks for a swap → HOD approves → timetable changes. If even more time is needed, TR-4 cuts makeup booking to "find slots only", which frees Taha.

The honest catch: after the swap, Amna's first backup is Taha, who has almost no spare time himself. In v1 her backup was the person with the most spare time. If both Amna and Taha slip, the backup is Sajawal, at the cost of his Should task and some of his 14 November bug-fixing time.

Part 7. Day-by-day overview and each person's page

Each person has about **2.6 hours of task work per day**. Hours in brackets are "likely" estimates.

7.1 Whole team, one table (end-of-day goal = what must exist at 21:00)

Day	Sajawal (backend)	Wajiha (backend)	Eisha (backend)	Amna (frontend)	Taha (frontend)	Team goal by 21:00
Sun 8	BE-01 Make the backend start (3)	BE-02 Fix import paths (1.5) · BE-03 Fix events and instructor name (1.5)	OP-01 Tests on every PR + template (2) · BE-06 Fix swaps, start (1)	FE-01 Frontend cleanup (3)	FE-03 Real login: API helper (2.5)	Backend starts, `/docs` shows all endpoints, CI runs, frontend is in `main`
Mon 9	BE-04 Fix login, start (2.5) · OP-02 First deploy help (0.5)	**BE-05 API contract, final (3)**	**OP-02 First deploy (2.5)** · BE-06 (0.5)	FE-04 Timetable page on mock data (3)	FE-02 Small JS fixes (0.75) · FE-03 Login on mocks (2)	**Contract final. Backend and frontend online. BE-06 branch pushed (TR-2).**
Tue 10	BE-04 merged by 13:00 (1.5) · BE-08 Safe makeup booking, start (1)	BE-10 Tables on MySQL (3)	**BE-06 merged, tests green (3.5)**	FE-04 (2.5)	FE-03 on real backend, merged (2) · FE-07 Bell, start (0.5)	Login works end to end. Tables exist online. Swaps fixed.
Wed 11	BE-08 merged (3)	BE-11 Sample data: core data by 18:00 (3)	BE-07 HOD approval (3)	**FE-04 on real data, merged (0.5)** · FE-05 Swap page, teacher half (2)	FE-06 Makeup page: find slots (3)	Timetable shows real sample data
Thu 12	OP-03 Full deploy (3)	BE-11 demo extras (1) · OP-04 README, start (1.5)	BE-09 Notification tests (2)	FE-05 teacher half merged (3)	FE-06 booking, merged (2) · FE-07 (0.5)	No fake data left. Staging has real data.
Fri 13	BE-12 Login attempt limit (1) · demo test on staging	OP-04 README merged (0.5) · BE-13 Roll-number format (0.5)	BE-14 File content check (1) · two-tab booking test	FE-05 HOD half merged (1) · FE-08 Secure link + image (0.35)	FE-07 Bell merged (2) · FE-09 Year list (0.25)	**Full demo passes once on staging**
Sat 14	FREEZE · rehearse ×2 · tag `v1.0`	Bug fixes · run the demo as an outsider	Bug fixes	Bug fixes	Bug fixes	Demo under 5 minutes, twice, no errors
Sun 15	Hand in · record backup demo video	Spare	Spare	Spare	Spare	Submitted

7.2 Sajawal: backend lead, plan owner (Must 14.5 h + Should 1 h, limit 15.6 h)

- **Your work is reviewed by:** Wajiha (main), Eisha (backup).
- **You review:** Wajiha's PRs, Eisha's BE-06 and OP-02, the API contract, and frontend PRs as backup. Keep to the 4-hour limit.
- **You own:** `main.py`, `requirements.txt`.
- **Your backup if you're stuck:** Wajiha (backend), Eisha (deploy).
- **You are also the backup for:** BE-06 and OP-02 (TR-2), and FE-03 (RK-9).

Day	Task	What exactly you do	Done when	Needs first
Sun 8	**BE-01 Make the backend start** (3 h). Covers BR-C1 (empty `main.py`), BR-C2 (missing settings file), BR-C3 (missing login check), BR-D3 (role trusted from the token).	Fill `main.py`: create the app, connect all 11 parts, allow the frontend website, add `/api/health`. Create `config.py`, which reads settings from `.env` with no default password. Create `dependencies.py` with the "who is logged in" check; it loads the user **from the database** every time, not from the token. Define the dummy password hash. Add `.env.example` with names only.	`uvicorn app.main:app` starts, `/api/health` says OK, and `/docs` lists every endpoint. PR merged tonight.	Nothing
Mon 9	**BE-04 Fix login and user accounts** (2.5 h). Covers BR-C5 (login doesn't match the user table), BR-M8 (open self-signup), BR-D6 (duplicate user crashes), FR-C1 (broken signup page; the page is removed). · **OP-02 First deploy** (0.5 h, with Eisha)	Add a roll-number column; make email optional; allow "no password yet" for un-activated students. Fix login, activate, "who am I" and admin create-user. Remove open self-signup. Return 409 for duplicates. Write login tests. Give Eisha the hosting accounts and check her first deploy.	Deploy: both websites answer online.	BE-01
Tue 10	**BE-04** finish (1.5 h) · **BE-08 Make makeup booking safe**, start (1 h)	Merge BE-04 **by 13:00**: Wajiha's BE-10 and Taha's FE-03 wait for it.	Login tests green; a wrong password gets 401.	BE-01
Wed 11	**BE-08 Make makeup booking safe** (3 h). Covers BR-M3 (lecture courses put in labs), BR-M4 (no ownership check or lock).	Lecture courses get lecture rooms only. Teachers can book only for themselves; HOD/admin for their department. Lock the room while saving. Add a database rule that refuses a duplicate room/date/time. Write tests.	Tests show a lab is refused for a lecture course, and a second booking of the same slot is refused.	BE-01
Thu 12	**OP-03 Full deploy** (3 h)	On Render: create tables on every deploy, load sample data on staging, fix the Python version, use the database's secure connection, set the health check, and write a smoke-test list.	Staging shows sample data after a fresh deploy.	BE-10, BE-11, OP-02
Fri 13	**BE-12 Limit login attempts** (Should, 1 h). Covers BR-D4 (no login rate limit). · Demo test on staging	5 attempts per minute per roll number, then 429. Run the Part 8 demo on the public website.	The 6th quick wrong attempt gets 429. The demo passes once.	BE-04
Sat 14	Freeze	Bug fixes only. Two timed rehearsals. Tag the release `v1.0`.	Demo under 5 minutes twice with no error.	n/a
Sun 15	Hand in	Record a backup demo video. Redeploy only for a real bug. Submit.	Submitted.	n/a

Every night at 21:00: post the critical-path line and any trigger that fired.

7.3 Wajiha: backend, import and portal (Must 15 h + Should 0.5 h, limit 15.6 h)

- **Your work is reviewed by:** Sajawal (main), Eisha (backup).
- **You review:** Sajawal's PRs (main), and Eisha's BE-07, BE-09 and OP-01 (main).
- **You own:** database migration files. Nobody else creates them.
- **Your backup if you're stuck:** Sajawal.
- **You are also the backup for:** Eisha's BE-07, BE-09 and OP-01.

Day	Task	What exactly you do	Done when	Needs first
Sun 8	**BE-02 Fix broken import paths and package files** (1.5 h). Covers BR-C4 (wrong import path in 12 files), BR-N2 (missing `__init__.py`), BR-D8 (uploads not ignored by Git). · **BE-03 Fix events update and instructor name** (1.5 h). Covers BR-M1 (events uses PATCH), BR-M2 (instructor name ignored).	Change `from database import` to `from app.database import` in 12 files. Add `pytest.ini`. Add empty `__init__.py` files. Add uploads, `*.db` and `node_modules` to `.gitignore`. Events update uses PUT. `instructor_name` uses the given name, else the uploader's, with one test per case.	`pytest` runs and the portal tests pass. Both PRs merged.	BE-01 (rebase on it)
Mon 9	**BE-05 Write the single API contract** (3 h). Covers BR-M9 (no single contract), FR-C7 (frontend sends email, backend wants roll number).	Create `docs/api-contract.md`: your file plus login, swaps, makeup, notifications and health. For each endpoint: address, who may call it, what you send, what comes back, error codes. Fix shared formats: login with **roll number**, day names `Monday…Sunday`, times `HH:MM`, dates `YYYY-MM-DD`, the four roles.	Reviewed by Sajawal **and** Taha, merged **by 21:00**. From now on, changing it needs both.	BE-01
Tue 10	**BE-10 Create database tables on MySQL** (3 h). Covers BR-M7 (no database migrations).	Set up Alembic in the morning. After BE-04 merges (about 13:00), generate the first migration and run it on the online Aiven database.	All tables exist on Aiven. If not by 21:00, TR-5 fires.	BE-04
Wed 11	**BE-11 Fill the database with sample data** (3 h)	Write `backend/scripts/seed.py`: SE department, ~8 rooms with real capacities (2 labs), ~6 courses, 2 sections, 4 teachers, a full-week timetable, one account per role, one un-activated student. Running it twice must not create duplicates.	**Core data on staging by 18:00**, because Amna needs it for FE-04.	BE-10
Thu 12	**BE-11** demo extras (1 h) · **OP-04 Write the README**, start (1.5 h). Covers BR-N9 (no backend README), FR-N9 (no frontend README), FR-M7 (scope decision).	Add a known clashing slot and a known free slot for the demo. README: local run steps for backend and frontend, settings, what is not built yet. Write down the scope decision: the portal pages stay in this repo; the timetable pages are the core.	Seed extras merged.	BE-11, OP-03
Fri 13	**OP-04** finish (0.5 h) · **BE-13 Use the real roll-number format** (Should, 0.5 h). Covers BR-N7 (roll-number pattern is a guess).	Add the deploy steps and demo accounts to the README. Fix the roll-number pattern using the real roll numbers Sajawal sends you, with 3–4 real examples as tests.	A stranger can run the project from the README.	OP-03
Sat 14	Freeze	Bug fixes only. Run the demo as if you'd never seen it; report each bug to its owner. Check the sample data loads cleanly twice.	n/a	n/a
Sun 15	Spare	Emergencies only.	n/a	n/a

7.4 Eisha: backend, swaps, HOD approval, notifications, tests on GitHub (Must 14.5 h + Should 1 h, limit 15.6 h)

- **Your work is reviewed by:** Sajawal for BE-06 and OP-02; Wajiha for BE-07, BE-09, OP-01 and BE-14.
- **You review:** Sajawal's and Wajiha's PRs as backup.
- **Your backup if you're stuck:** Sajawal (BE-06, OP-02), Wajiha (everything else).
- **Important:** BE-06 is on the critical path, and you have no commits in the repo yet. **Push your branch every day, even unfinished.** If no BE-06 branch exists by 9 Nov 21:00, TR-2 moves BE-06 to Sajawal and gives you OP-03 (full deploy). That is not a punishment; it keeps the team on time. Ask for help early.

Day	Task	What exactly you do	Done when	Needs first
Sun 8	**OP-01 Automatic tests on every PR and a PR template** (2 h). Covers BR-C7 (code merged without ever being run). · **BE-06 Fix the swap feature and conflict checker**, start (1 h).	Add `.github/workflows/backend-tests.yml` (installs and runs `pytest` on every PR) and `.github/pull_request_template.md` (checklist from the guide). Then clone the repo, run the backend, and read `swap_service.py` and `conflict_checker.py`.	CI runs on new PRs (red is OK until BE-02 merges).	Nothing
Mon 9	**OP-02 First deploy** (2.5 h, Sajawal helps 0.5 h) · **BE-06** (0.5 h)	Backend on Render from `main`, database on Aiven, frontend on Netlify. Settings go on the host, never in code. Allow the Netlify address (CORS). Check the free plans still exist. Rename every `TimetableEntry.day` to `day_of_week` and **push the branch**.	Both public websites answer; `/api/health` says OK online. **BE-06 branch pushed by 21:00 (TR-2).**	BE-01
Tue 10	**BE-06 Fix the swap feature and conflict checker** (3.5 h). Covers BR-C6 (swaps use a column that doesn't exist), BR-D1 (no tests for swaps and the conflict checker).	Give `TimetableEntry` a link to its teacher. Write swap tests and conflict-checker tests: one success, one permission refusal, one invalid state per endpoint.	**Merged with green tests by 21:00** (TR-3).	BE-04
Wed 11	**BE-07 Let each HOD approve their own department's swaps** (3 h). Covers BR-D2 (HOD accounts can never approve), BR-D1 (no permission tests).	Admin create-user takes a department, required for HODs. Tests: HOD of department A can approve A's swap and is refused for B's; a student can't approve.	Merged, tests green.	BE-06
Thu 12	**BE-09 Test notifications** (2 h). Covers BR-D1 (no notification tests).	Test that a notification is saved when a swap is approved and when a makeup class is booked, and that unread-count and mark-read work.	Merged, tests green.	BE-06, BE-08
Fri 13	**BE-14 Check uploaded files by content** (Should, 1 h). Covers BR-D7 (uploads checked by extension only). · Two-tab booking test	Check the first bytes of PDFs and images before saving. On staging, try to approve two clashing bookings at the same moment in two tabs.	A fake `.pdf` is refused; the second booking is refused.	BE-08
Sat 14	Freeze	Bug fixes only.	n/a	n/a

Day	Task	What exactly you do	Done when	Needs first
Sun 15	Spare	Emergencies only.	n/a	n/a

7.5 Amna: frontend, timetable screens (Must 15 h + Should 0.35 h, limit 15.6 h)

- **Your work is reviewed by:** Taha (main), Sajawal (backup).
- **You review:** Taha's PRs (main).
- **Your backup if you're stuck:** Taha, then Sajawal.
- **Read Part 6b:** it says exactly what happens if you fall behind. Tell the team early. That is not failure.

Day	Task	What exactly you do	Done when	Needs first
Sun 8	**FE-01 Clean up the frontend and bring it into `main`** (3 h). Covers FR-C1 (signup page broken), FR-C5 (forgot password unreachable), FR-C6 (lecture page crashes), FR-M1 (pages without login guard), FR-M2 (page list doesn't match file names), FR-M3 (dead links), FR-M4 (duplicate folder), FR-D4 (unused CSS files), FR-N1 (file name typo).	New branch from the latest `main`; copy in `frontend/`. Delete the `web project frontend/` folder and the 4 unused CSS files. Rename `forgetpassword.html` to `forgot-password.html`. Make the `PAGES` list match the real file names and fix the dead links. Load `common.js` on every page. Rebuild `lecture.html` so it loads `lecture.js`. Delete the signup page (students activate instead). Label the upload buttons "demo only" (FR-M5 stays deferred). Title the PR honestly: "Frontend UI, mock data".	Every page opens with no red errors in the browser console; logged-out users are sent to login. PR merged tonight.	Nothing
Mon 9	**FE-04 Timetable page** (3 h)	`timetable.html`: a weekly view. Students see their section, teachers their own classes, and makeup classes show too. Use mock data shaped like the contract.	The page shows the mock week correctly.	BE-05 (draft)
Tue 10	**FE-04** (2.5 h)	Finish the grid on mock data. Use Taha's `NUML.api()` helper.	n/a	FE-03 helper
Wed 11	**FE-04** on real data (0.5 h) · **FE-05 Swap requests page**, teacher half (2 h)	Switch the timetable to the real API once the core sample data is on staging (18:00). Start the teacher swap form.	**FE-04 merged by 21:00**, showing real sample data. If not, TR-8 fires.	BE-11, FE-03
Thu 12	**FE-05 teacher half** (3 h)	The teacher picks a class and a new slot. On a clash, show the 409 reason; on success, show "pending".	Teacher half merged; a request appears as pending.	BE-06
Fri 13	**FE-05 HOD half** (1 h) · **FE-08 Secure software link and faster hero image** (Should, 0.35 h). Covers FR-N11 (MARS link uses http), FR-N8 (large image with no preload).	An HOD/admin list with approve and reject buttons. Then change the MARS link to https and preload the hero image.	The HOD approves on staging and the teacher's timetable changes.	BE-07
Sat 14	Freeze	Bug fixes only.	n/a	n/a
Sun 15	Spare	Emergencies only.	n/a	n/a

7.6 Taha: frontend, login, makeup page, notification bell (Must 15.25 h + Should 0.25 h, limit 15.6 h)

- **Your work is reviewed by:** Amna (main), Sajawal (backup).
- **You review:** Amna's PRs (main), and the API contract with Sajawal.
- **You own:** `common.js` and `api.js` (after FE-01 merges). Others ask you for changes.
- **Your backup if you're stuck:** Sajawal (for FE-03), then Amna.
- **You are Amna's first backup** (TR-8).
- **Important:** FE-03 blocks every other frontend page. Your earlier login code decided login in the browser. This time **the server decides**. Push daily.

Day	Task	What exactly you do	Done when	Needs first
Sun 8	**FE-03 Real login on the frontend**: API helper (2.5 h)	Start `js/api.js` with one `API_BASE` setting (empty = use mock files) and `NUML.api(path, options)`, which sends the login ticket with every request and goes to login on 401. Share it with Amna tonight.	Amna can load mock data through `NUML.api()`.	Nothing
Mon 9	**FE-02 Small JavaScript fixes** (0.75 h). Covers FR-D8 (forgot password pretends to send a link), FR-N2 (wrong comment about `defer`), FR-N3 (dead code in `dashboard.js`), FR-N4 (duplicate logout redirect). · **FE-03** login on mocks (2 h)	After FE-01 merges: the forgot-password page says "Contact your department admin"; fix the comment; delete the dead code and the extra redirect. Build the login form with **roll number**, not email.	FE-02 merged.	FE-01
Tue 10	**FE-03 Real login on the frontend** (2 h). Covers FR-C2 (any password accepted), FR-C3 (login is only a browser flag), FR-C4 (no backend calls), FR-C7 (email vs roll number), FR-D1 (personal data kept in the browser), FR-D7 (inline form handlers). · **FE-07 Notification bell**, start (0.5 h)	Connect to the real backend after BE-04 merges (13:00). Check "who am I" on page load. Logout clears the ticket. Send each role to its start page. Delete the saved profiles in the browser. Replace `onsubmit=` with `addEventListener`. Start the bell on the real unread-count endpoint.	**FE-03 merged by 21:00.** A wrong password is refused; a fake "logged in" value in the browser no longer works.	BE-04, BE-05
Wed 11	**FE-06 Makeup class page** (3 h)	`makeup.html`: pick a section and a duration → show free slots and rooms from the real find-slots endpoint.	Slots appear for the sample section.	FE-03
Thu 12	**FE-06** booking (2 h) · **FE-07** (0.5 h)	Book a slot. Show the 409 message if the slot was taken.	**FE-06 merged:** a teacher books a makeup on staging and it shows on the timetable.	BE-08
Fri 13	**FE-07 Notification bell** (2 h) · **FE-09 Sensible year list on the lectures page** (Should, 0.25 h). Covers FR-N6 (years listed back to 2000).	The bell asks for the unread count every 15 seconds and shows a dropdown with mark-as-read. Then set the first year to 2018.	The bell shows "swap approved" after an HOD approves.	BE-09
Sat 14	Freeze	Bug fixes only.	n/a	n/a
Sun 15	Spare	Emergencies only.	n/a	n/a

Part 8. The 5-minute demo (everything above is built for this)

Two minutes before: open `/api/health` to wake the server. Each person presents the part they built, so everyone can answer the instructor's questions about it.

Time	Step	Presenter
0:00	Open the public website → login page. Mention `/docs` is live.	Sajawal
0:20	**Student** logs in with a roll number → sees their section's weekly timetable.	Amna
1:00	**Teacher** logs in → own timetable → asks to move a class to a clashing slot → red message, for example "Room R-2 is booked" → picks a free slot → request is **pending**.	Amna
2:00	**HOD** logs in → swap requests → approves. Explains the HOD-only-own-department rule.	Eisha
2:30	**Teacher** → bell shows "Swap approved" → the class is in its new slot.	Taha
3:00	**Teacher** → makeup page: section + 90 minutes → free slots in lecture rooms only → books one. Sajawal explains how the solver picks rooms.	Taha, Sajawal
3:45	**Student** → bell shows "Makeup class added" → it's on the timetable.	Amna
4:15	**Admin** → `/docs`: timetable import (test run) and the portal endpoints (exams, GPA) → end.	Wajiha

The sample data guarantees fixed accounts, one known clashing slot, one known free slot, and a section with enough room capacity.

Part 9. What I need from you (Sajawal) in the next 2 hours

1. **Post Part 4 (rules) and Part 7 (pages) in the team chat tonight.** State the swap clearly: Taha is frontend, Eisha is backend. Get a written "yes, I own it" from **Eisha (BE-06 and OP-02)** and **Taha (FE-03)**. No reply by morning means RK-1 or RK-9 is happening.
2. **Check Eisha can work:** she has write access (`Eishashahzad`) but has never pushed. Ask her to clone the repo and push one small branch tonight.
3. **Create the Render, Aiven and Netlify accounts** and add Eisha. Confirm the free plans still exist. Without them, OP-02 can't happen on Monday.
4. **Start BE-01 (make the backend start) tonight or first thing tomorrow.** The whole critical path waits for it.
5. **Send Wajiha the real SE timetable sheet and 3–4 real roll numbers** (for BE-11 and BE-13).

Appendix: technical version (for Sajawal)

A1. Audit evidence (unchanged from v1)

- `backend/app/main.py` is 0 bytes, so 0 of 53 coded endpoints are registered. I ran `pytest` in a Python 3.13 scratch venv: `ModuleNotFoundError: app.config` at `conftest.py:19`, 0 tests run. The 112/116 figure is from the backend review's stubbed run; I didn't reproduce it.
- PRs: 0 open. Merged backend PRs: #27, #28, #30, #26, #33, #34. Branches: `main` (protected), `frontend` (6 ahead, 6 behind; PR Sajawal3185/timetable-automation#35 closed unmerged 6 Nov), `amna_245` (stale since 17 Oct). No commits between 24 Oct and 7 Nov.
- Tests use in-memory SQLite (`conftest.py:44`), where `with_for_update()` does nothing, so locking is untested.
- **Track records after the swap** (git history on every branch + GitHub collaborators):
  - Sajawal: 21 commits on main (PRs #27, #28, #30, #33, #34; core, auth, makeup, swaps, notifications).
  - Wajiha: PRs #15 and #26 (portal modules, import, 5 test files).
  - Amna: PRs #14 and #16, plus `frontend` commits 16–24 Oct.
  - **Taha (`Taha-ops116`, now frontend):** commits `5f040aa` (21 Oct, "Authentication and main screen") and `8471ea6` (24 Oct, "Integration done") on `frontend`; co-author on `84e3db9` and `7618b2e`. 0 tests, 0 fetch/XHR calls.
  - **Eisha (`Eishashahzad`, now backend):** write access, 0 commits, 0 PRs, 0 tests.

A2. Task estimates (best / likely / worst hours)

Expected time = (best + 4 × likely + worst) ÷ 6. Capacity uses the likely value.

Task	Best/Likely/Worst	Depends on
BE-01 Make the backend start	2/3/4	none
BE-02 Import paths	1/1.5/2	none
BE-03 Events + instructor name	1/1.5/2	BE-02
BE-04 Login fix	3/4/6	BE-01
BE-05 API contract	2/3/4	BE-01
BE-06 Swaps fix	3/5/8	BE-01, BE-04
BE-07 HOD approval + permission tests	2/3/4	BE-06
BE-08 Safe makeup booking	3/4/6	BE-01
BE-09 Notification tests	1/2/3	BE-06, BE-08
BE-10 Migrations on MySQL	2/3/5	BE-04
BE-11 Sample data	3/4/6	BE-10
BE-12 Login attempt limit	0.5/1/1.5	BE-04
BE-13 Roll-number format	0.25/0.5/1	real roll numbers
BE-14 File content check	0.5/1/1.5	BE-02
BE-15 Swap vs makeup check	2/3/5	BE-06, BE-08
BE-16 One role-check system	1.5/2.5/4	BE-07
BE-17 Service transactions	1/2/3	BE-06
BE-18 Unknown capacity	0.5/1/1.5	BE-02

Task	Best/Likely/Worst	Depends on
BE-19 Term from settings	0.5/1/2	BE-01
BE-20 Backend tidy-up	2/3.25/4	none
BE-21 Real AI/CS/IT sheets	2/3/6	BE-10
FE-01 Frontend cleanup	2/3/5	none
FE-02 Small JS fixes	0.5/0.75/1	FE-01
FE-03 Real login (frontend)	4/6.5/9	BE-05 (mock), BE-04 (real)
FE-04 Timetable page	4/6/9	FE-03, BE-11
FE-05 Swap requests page	4/6/9	FE-04, BE-06, BE-07
FE-06 Makeup page	3/5/7	FE-03, BE-08
FE-07 Notification bell	2/3/4	FE-03, BE-09
FE-08 Secure link + preload	0.25/0.35/0.5	FE-01
FE-09 Year list	0.1/0.25/0.5	FE-01
FE-10 Shared layout	1/2/3	FE-01
FE-11 Activate page	1/1.5/2	FE-03
FE-12 Empty rooms page	2/3/4	FE-03
FE-13 Papers + slides on real data	4/5/8	FE-03
FE-14 GPA page	2/3/4	FE-03
FE-15 Software + events pages	2/3/4	FE-03
FE-16 Font + linter	1/1.5/2	FE-01
FE-17 Accessibility	1/1.5/3	FE-01
OP-01 CI + PR template	1/2/3	none
OP-02 First deploy	2/3/5	BE-01
OP-03 Full deploy	2/3/5	BE-10, BE-11, OP-02
OP-04 README	1.5/2/3	OP-03
OP-05 Postman	1/1.5/2	BE-11
OP-06 MySQL in CI	1/2/3	OP-01, BE-10

The critical path BE-01 → BE-04 → BE-10 → BE-11 → FE-04 → FE-05 → OP-03 is about 23 likely hours, run in sequence over 6 days. It works only with same-day handoffs.

A3. Traceability: every review item and where it went

Status: **Must**, **Should** = scheduled; **Rule** = process rule; **Deferred** = Could, after release.

Backend review (35 items)

Item	Problem	Status	Task
BR-C1	`main.py` is empty	Must	BE-01
BR-C2	`config.py` missing	Must	BE-01
BR-C3	Login check (`get_current_user`) missing	Must	BE-01
BR-C4	Wrong import path in 12 files	Must	BE-02

Item	Problem	Status	Task
BR-C5	Login code doesn't match the user table	Must	BE-04
BR-C6	Swaps use a missing column and relationship	Must	BE-06
BR-C7	Code merged without being run	Must + Rule	OP-01, Rule 3
BR-M1	Events update uses PATCH, contract says PUT	Must	BE-03
BR-M2	`instructor_name` ignored for teachers	Must	BE-03
BR-M3	Lecture courses put into labs	Must	BE-08
BR-M4	Makeup booking: no ownership check or lock	Must	BE-08
BR-M5	Two rule engines disagree (swaps ignore makeups)	Deferred	BE-15
BR-M6	Four role-check implementations	Deferred	BE-16
BR-M7	No database migrations	Must	BE-10
BR-M8	Open self-registration	Must	BE-04
BR-M9	No single API contract	Must	BE-05
BR-D1	Riskiest modules have no tests	Must	BE-04 (auth), BE-06 (swaps, conflict checker), BE-07 (permissions), BE-09 (notifications)
BR-D2	HODs can never approve swaps	Must	BE-07
BR-D3	Role trusted from the token	Must	BE-01 (merged with the user-loading fix)
BR-D4	No login rate limit	Should	BE-12
BR-D5	Placeholder room capacity treated as real	Deferred	BE-18
BR-D6	Duplicate user gives 500	Must	BE-04
BR-D7	Uploads checked by extension only	Should	BE-14
BR-D8	`uploads/` not in `.gitignore`	Must	BE-02
BR-D9	Services roll back the caller's session	Deferred	BE-17
BR-N1	Dev tools in production requirements	Deferred	BE-20
BR-N2	Missing `__init__.py` files	Must	BE-02
BR-N3	Test folder name (`test/` vs `tests/`)	Deferred	BE-20
BR-N4	People's names in code and file names	Deferred	BE-20
BR-N5	Stale comments	Deferred	BE-16
BR-N6	Hardcoded term "Fall 2026"	Deferred	BE-19
BR-N7	Roll-number pattern is a guess	Should	BE-13
BR-N8	Proposal `.docx` in repo root	Deferred	BE-20
BR-N9	No backend README	Must	OP-04
BR-N10	No linter or formatter	Deferred	BE-20

Frontend review (35 items)

Item	Problem	Status	Task
FR-C1	Signup page broken	Must	FE-01 (page removed), BE-04 (self-signup removed)
FR-C2	Login accepts any password	Must	FE-03
FR-C3	Auth is a browser-only flag	Must	FE-03
FR-C4	No backend calls despite the "Integration done" commit	Must + Rule	FE-03, Rule 3 (honest PR titles)
FR-C5	Forgot password page unreachable	Must	FE-01
FR-C6	Lecture page crashes	Must	FE-01
FR-C7	Email vs roll number mismatch	Must	BE-05, FE-03
FR-M1	Pages without login guard	Must	FE-01
FR-M2	Page list doesn't match file names	Must	FE-01
FR-M3	Links to pages that don't exist	Must	FE-01
FR-M4	Duplicate `web project frontend/` folder	Must	FE-01
FR-M5	Uploads are fake and vanish	Deferred	FE-13 (upload buttons labelled "demo only" in FE-01 meanwhile)
FR-M6	Lecture data points at missing files	Deferred	FE-13
FR-M7	Scope drift (archive vs timetable)	Must	OP-04 (decision written in README)
FR-M8	Duplicate subject list in two files	Deferred	FE-13
FR-D1	Profiles kept in browser storage	Must	FE-03
FR-D2	No shared layout	Deferred	FE-10
FR-D3	Nav styles injected from JavaScript	Deferred	FE-10
FR-D4	Four unused CSS files	Must	FE-01
FR-D5	`innerHTML` with template values in GPA page	Deferred	FE-14
FR-D6	Password rule shown wrong (6 vs 8)	Deferred	FE-11 (signup page deleted in FE-01; the activate page uses 8)
FR-D7	Inline `onsubmit=` handlers	Must	FE-03
FR-D8	Forgot password does nothing	Must	FE-02 (message); a real reset is Won't
FR-D9	Fonts differ by page	Deferred	FE-16
FR-N1	File name typo `forgetpassword.html`	Must	FE-01
FR-N2	Wrong comment about `defer`	Must	FE-02
FR-N3	Dead code in `dashboard.js`	Must	FE-02
FR-N4	Duplicate logout redirect	Must	FE-02
FR-N5	Commit hygiene	Rule	Squash merge (guide rule)
FR-N6	Year list back to 2000	Should	FE-09
FR-N7	Accessibility gaps	Deferred	FE-17
FR-N8	Large image with no preload	Should	FE-08
FR-N9	No frontend README	Must	OP-04
FR-N10	No linter config	Deferred	FE-16

Item	Problem	Status	Task
FR-N11	MARS link uses http	Should	FE-08

Count: 70 items. **48 scheduled** (42 Must, 6 Should; 2 of the Musts also have a rule). **1 rule only** (FR-N5). **21 deferred**, each with its reason in Part 5.4. **None dropped.**

A4. Interface details pinned on day 1 (in BE-05, the API contract)

- `{API_BASE}/api/...`, JSON except multipart uploads. `Authorization: Bearer <JWT>`, 60-minute expiry, no refresh. Errors are `{"detail": "..."}`.
- Roles: `student|teacher|hod|admin`. Days: `Monday…Sunday`. Swap status: `pending|approved|rejected|cancelled`. Times `HH:MM`, dates `YYYY-MM-DD`.
- First 5 endpoints:

  1. `POST /api/auth/login` (`auth.py:173`, fixed by BE-04)
  2. `GET /api/auth/me` (`auth.py:227`, fixed by BE-01 and BE-04)
  3. `GET /api/timetable` (`timetable.py:31`, fixed by BE-02)
  4. `POST/GET /api/swap-requests` and `/{id}/approve` (`swaps.py:32,61,72`, fixed by BE-06)
  5. `POST /api/makeup/find-slots` and `POST /api/makeup` (`makeup.py:35,61`, fixed by BE-08)

  Plus `GET /api/notifications/unread-count` (exists) and `GET /api/health` (new, added in BE-01).

A5. Deployment configuration

- **Render:** root `backend/`; build `pip install -r requirements.txt`; start `alembic upgrade head && uvicorn app.main:app --host 0.0.0.0 --port $PORT`; `PYTHON_VERSION=3.12`.
- **Database:** Aiven free MySQL; fallback TiDB Serverless, then Postgres.
- **Frontend:** Netlify serving `frontend/`; a `config.js` per environment sets `API_BASE`.
- **Host environment variables:** `DATABASE_URL=mysql+pymysql://…?ssl_ca=…`, `JWT_SECRET_KEY`, `CORS_ORIGINS=https://<site>`, `UPLOAD_DIR`, `ACCESS_TOKEN_EXPIRE_MINUTES=60`.

Failure modes:

1. CORS origin mismatch.
2. MySQL TLS not passed to PyMySQL.
3. Constraint names over 64 characters (from the naming convention).
4. Enum DDL.
5. Cold start.
6. Temporary disk wipes uploads.
7. A missing environment variable crashes startup.
8. Python 3.13 vs pinned wheels.

Change log (v1 → v2)

Merged in from the two reviews

- All 70 review items were added as tasks or listed as deferred (traceability table A3).
- Folded into existing tasks, not duplicated:
  - BR-D3 into BE-01 (the user-loading fix).
  - BR-C5, BR-M8 and BR-D6 into BE-04.
  - FR-C1 into FE-01 + BE-04 (the signup page and self-signup are both removed).
  - FR-C7 into BE-05 + FE-03.
  - The BR-D1 test gaps split across BE-04, BE-06, BE-07 and BE-09.
  - FR-M7 into OP-04.
- New tasks created from the reviews:
  - FE-02 (FR-D8, FR-N2, FR-N3, FR-N4)
  - BE-12 (BR-D4), BE-13 (BR-N7), BE-14 (BR-D7)
  - FE-08 (FR-N8, FR-N11), FE-09 (FR-N6)
  - Could tasks BE-15 to BE-20, FE-14, FE-16, FE-17
- BE-07 grew from 2 to 3 hours (permission tests). FE-01 grew to cover FR-N1 and FR-D4. FE-03 grew from 6 to 6.5 hours (FR-D7). OP-04 grew from 1.5 to 2 hours (frontend README + scope decision).
- **Removed:** the separate review documents from the pack. This plan is the only document.

Code scheme

- One scheme: BE-, FE-, OP- for tasks; BR-, FR- for review sources; TR- for triggers; RK- for risks.
- Old codes replaced:
  - B-1…B-12 → BE-01…BE-11 and OP-01 (B-12)
  - D-1, D-2, D-3 → OP-02, OP-03, OP-04
  - F-1…F-7 → FE-01, FE-03…FE-07; the old shared layout became FE-10 (Could)
  - Old Should/Could codes → BE-, FE- and OP- tasks, with priority shown as a column. The old X-codes map as follows:
    - Postman → OP-05
    - Swap vs makeup → BE-15
    - Empty rooms → FE-12
    - Papers/slides → FE-13
    - Activate → FE-11
    - Safety fixes → BE-12, BE-14, BE-18
    - MySQL in CI → OP-06
    - Real sheets → BE-21
  - T1…T7 → TR-1, TR-3…TR-8 (TR-2 is new); R1…R8 → RK-1…RK-8
- Review severities shown as C, M, D, N (minor renamed to N so it doesn't clash with M).

Taha / Eisha swap: every place changed

- "What is new" and the Part 1 risk line.
- Glossary rows "Backend" and "Frontend" (who builds them).
- Rule 3: reviewer table rebuilt (backend reviewed by backend, frontend by frontend; Sajawal is the frontend backup reviewer).
- Rule 4: `common.js` and `api.js` owner changed from Eisha to Taha.
- Rule 6: TR-1 (no longer "Eisha's spare time"), TR-2 (new, Eisha → Sajawal), TR-3 (Taha → Eisha in the condition; the exchange now gives Eisha OP-03), TR-6 backups, TR-8 (Amna's backup changed from Eisha to Taha, and what it costs).
- Part 5.3: the equal-division table.

- Part 5.4: the explainer column (portal screens → Taha; swap vs makeup and notifications → Eisha).
- Part 6:
  - RK-1 rewritten for Eisha (backend, 0 commits).
  - RK-9 new for Taha (frontend login).
  - RK-10 new: no frontend slack.
  - 6b: Amna's backup is now Taha.
- Part 7: 7.1 columns reordered by role. The Eisha and Taha pages were rebuilt (owner, reviewers, backups, tasks). Sajawal's, Wajiha's and Amna's pages had their reviewer and backup lines updated.
- Part 8: demo presenters added by the new roles.
- Part 9: confirm Eisha (BE-06, OP-02) and Taha (FE-03); check Eisha can push; add Eisha (not Taha) to the hosting accounts.
- Appendix A1: track records rewritten. A2: owners updated.

What moved between people (to equalise hours)

- README (OP-04): Sajawal → Wajiha.
- First deploy (OP-02): Sajawal's share 1 h → 0.5 h; Eisha 2.5 h.
- CI (OP-01): Eisha-as-frontend (v1) → Eisha as backend (kept, now role-consistent).
- Makeup page (FE-06): the backend person (v1 Taha) → Taha as frontend (kept with Taha).
- Shared layout (FE-10): v1 Eisha → Could, owner Taha.

What moved between priorities

- Must → Could: shared layout (FE-10), with reason in Part 5.4.
- Should → Could: Postman (OP-05), swap vs makeup (BE-15), empty rooms (FE-12), papers/slides (FE-13), activate page (FE-11), unknown capacity (BE-18), MySQL in CI (OP-06), real sheets (BE-21). Reason: no feature hours before the freeze.
- Should kept, now with owners and days: BE-12, BE-13, BE-14, FE-08, FE-09.

Corrections

- Capacity: feature hours are 78, not 91, because 14 Nov is freeze day.
- v1's "about 75 h" of Musts was exactly 73.5 h.
- Milestone "frontend on real APIs, 12 Nov" now holds only partly (Part 5.6).